

MAT425 review

SUNAY JOSHI

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§0.1 Cast of characters

Riemann, Cantor, Lebesgue, Borel, Cantelli, Caratheodory, Fatou, Markov, Hilbert, Riesz, Radon, Nikodym, von Neumann, Besicovitch, Hölder, Lipschitz

§0.2 Measures

add theorems...

[add proof sketches...]

§0.3 Lebesgue integral

Semiring: closed under intersections and set difference?...

Outer measure...

Caratheodory measurables...

Caratheodory extension...

Standard machine...

Thm (Egorov): $(X, \mathcal{A}, \mu)$ measure space, f_k measurable, $f_k \rightarrow f$ μ -a.e. in E . Then given $\varepsilon > 0$, there exists $E_\varepsilon \subseteq E$ such that $\mu(E \setminus E_\varepsilon) \leq \varepsilon$ and $f_k \rightarrow f$ uniformly on E_ε .

[Idea: consider a quantitative notion of “bad points”. Given δ, k , (k, δ) -bad points in E are those for which $f_{k'}(x)$ is more than δ away from $f(x)$ for some $k' \geq k$; “the (k, δ) -stragglers”. Let $A_{k, \delta} = \{x \in E : \exists k' \geq k, |f_{k'}(x) - f(x)| > \delta\}$ be this set. Since $f_k \rightarrow f$ a.e., $A_{k, \delta} \searrow \emptyset$ as $k \rightarrow \infty$. Since $\mu(E) < \infty$, $\mu(A_{k, \delta}) \searrow 0$. Let $\delta_i = \frac{1}{i}$ and k_i be chosen such that $\mu(A_{k_i, \delta_i}) \leq \frac{\varepsilon}{2^{i+1}}$. Let $E_\varepsilon = E \setminus \bigcup_i A_{k_i, \delta_i}$. Clearly $\mu(E \setminus E_\varepsilon) \leq \varepsilon$. If $x \in E_\varepsilon$, then for each i , past k_i , x is “good”. Thus $\sup_{E_\varepsilon} |f_k - f| \leq \delta_i$ for $k \geq k_i$. Thus $\limsup_k \sup_{E_\varepsilon} |f_k - f| \leq \delta_i$ for all i , so $\lim_k \sup_{E_\varepsilon} |f_k - f| = 0$.]

Thm (Lusin): $f : E \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ measurable, E has finite measure. Then given $\varepsilon > 0$, there exists $E_\varepsilon \subseteq E$ such that $\mathcal{L}^d(E \setminus E_\varepsilon) \leq \varepsilon$ and $f|_{E_\varepsilon}$ is continuous.

(Lusin is specific to Euclidean space.)

[Standard machine + Egorov.

If $f = \chi_F$ for $F \subseteq E$, think of the case $F = \text{interval}$ and $E = \mathbb{R}$. Construct $C \subseteq F$ closed and $C' \cap F = \emptyset$ closed such that $\mathcal{L}^d(F \setminus C), \mathcal{L}^d((F^c \setminus C') \cap E) \leq \frac{\varepsilon}{2}$. Set $E_\varepsilon = C \cup C'$. Since C, C' disjoint closed, $f|_{E_\varepsilon}$ is continuous.

If f is a simple function, do the previous construction for each of the n characteristic functions to precision $\frac{\varepsilon}{n}$ and throw away all “bad sets”.

If f is a measurable function, construct simple functions $f_k \rightarrow f$ a.e. on E . By previous part, if we throw away a “bad set” of size $\frac{\varepsilon}{2^{k+2}}$ for f_k for all k (hence a total measure of $\frac{\varepsilon}{2}$), then restricted to the remaining set, (f_k) are continuous. By **Egorov**, if we throw away another set of size $\frac{\varepsilon}{2}$, then (f_k) are continuous and $f_k \rightarrow f$ uniformly on E_ε . Thus f is continuous on E_ε .]

Borel-Cantelli, Fatou, MCT, DCT, partial converse to DCT...

Defn: $L^1(X, \mathcal{A}, \mu)$ (abbreviated $L^1(\mu)$) consists of all integrable f , modulo the equivalence relation $f \sim g$ if $f = g$ a.e.

Thm: L^1 is a complete normed vector space.

[To check vector space, use Minkowski: $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$. To check complete, take a Cauchy sequence (f_k) wrt L^1 norm. STS that a subsequence converges. Trick: “speed up” the rate. Abusing notation, extract a subsequence (f_k) such that $\|f_k - f_l\|_1 \leq \frac{1}{2^k}$ for all $l \geq k$. Our candidate limit is $f := f_1 + \sum_k (f_{k+1} - f_k)$. We choose this function because its partial sums are (f_k) , so that $f_k \rightarrow f$ pointwise a.e. f is well-defined because **by MCT**, $\int |f_1| + \sum_k |f_{k+1} - f_k| \leq \|f_1\|_1 + 2 < \infty$, so the integrand is finite a.e., so $f(x)$ converges absolutely for a.e. x . By the same computation, $f \in L^1$, since $\|f\|_1 \leq \|f_1\|_1 + 2 < \infty$. Since $f_k \rightarrow f$ pointwise a.e. and since $|f_k| \leq |f| \in L^1$, **by DCT** we have $f_k \xrightarrow{L^1} f \in L^1$, as desired.]

One can show the same for L^p , $p \geq 1$.

Thm (Holder): if $p \geq 1$, $\frac{1}{p} + \frac{1}{q} = 1$, then $\|fg\|_1 \leq \|f\|_p \|g\|_q$.

[Use Young’s inequality $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$, which follows from Jensen applied to $f(x) = \log x$. Assume $\|f\|_p = \|g\|_q = 1$. By Young, $|fg| \leq \frac{|f|^p}{p} + \frac{|g|^q}{q}$. Integrating, $\|fg\|_1 \leq \frac{1}{p} + \frac{1}{q} = 1 = \|f\|_p \|g\|_q$, as desired.]

Thm (Minkowski): if $p \geq 1$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ for all $f, g \in L^p$.

[Idea: start with LHS, split off a factor of $|f + g|$, apply triangle inequality to it, then use Holder on each term. Formally, $\int |f + g|^p \leq \int |f| |f + g|^{p-1} + \int |g| |f + g|^{p-1}$. By Holder, first term is at most $\|f\|_p \|f + g\|_p^{p/q}$. Thus $\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p/q}$, which implies the result.]

Density of simple functions in L^1 , continuous functions in $L^1(\mathbb{R}^d)$...

§0.4 Product measures

Defn: given σ -finite **complete** (!) measure spaces $(X, \mathcal{A}, \mu), (Y, \mathcal{B}, \nu)$, define $\mathcal{A} \otimes \mathcal{B} = \sigma(\mathcal{A} \times \mathcal{B})$. The product measure $\mu \otimes \nu$ is defined as $(\mu \otimes \nu)(E \times F) = \mu(E)\nu(F)$ on $\mathcal{A} \times \mathcal{B}$ and extends uniquely to $\mathcal{A} \otimes \mathcal{B}$ by Caratheodory. We typically extend $\mu \otimes \nu$ to the completion $\overline{\mathcal{A} \otimes \mathcal{B}}^{\mu \otimes \nu}$.

(To apply Caratheodory, must check that $\mathcal{A} \times \mathcal{B}$ is a semiring and $\mu \otimes \nu$ is σ -finite and σ -additive on $\mathcal{A} \times \mathcal{B}$, add...)

How to compute with $\mu \otimes \nu$? First, we give a formula for computing $(\mu \otimes \nu)(E)$; then, we give a formula for integrating $d(\mu \otimes \nu)$.

Defn: given $E \subseteq X \times Y$, its slice is $E_x = \{y \in Y : (x, y) \in E\}$.

Lem: for any $E \in \bigcap_{\sigma} \bigcup_{\sigma} \mathcal{A} \times \mathcal{B}$, we have

- E_x is $\mathcal{B}$ -measurable
- $x \mapsto \nu(E_x)$ is $\mathcal{A}$ -measurable
- $(\mu \otimes \nu)(E) = \int_X \nu(E_x) d\mu(x)$

[The choice of $\bigcap \bigcup \mathcal{A} \times \mathcal{B}$ is tricky! If $E = A \times B$ is a rectangle, everything is clear. Next, consider $E \in \bigcup \mathcal{A} \times \mathcal{B}$. Then $E = \bigsqcup_k E_k$, where E_k are rectangles, so that $E_x = \bigsqcup_k (E_k)_x \in \mathcal{B}$, $\nu(E_x) = \sum_k \nu((E_k)_x)$ is $\mathcal{A}$ -measurable, and

$$\int_X \nu(E_x) d\mu(x) = \int_X \sum_k \nu((E_k)_x) d\mu(x) \stackrel{MCT}{=} \sum_k (\mu \otimes \nu)(E_k) = (\mu \otimes \nu)(E) \quad (1)$$

Finally, consider $E \in \bigcap \bigcup \mathcal{A} \times \mathcal{B}$. Note $E \searrow E_k$ for some $E_k \in \bigcup \mathcal{A} \times \mathcal{B}$. Now everything follows from decreasing continuity, since $E_x \searrow (E_k)_x$ and **using finiteness!**]

We now extend to $\overline{\mathcal{A} \otimes \mathcal{B}}$.

Lem: for any $E \in \overline{\mathcal{A} \otimes \mathcal{B}}$, we have

- E_x for $\mathcal{B}$ -measurable for μ -a.e. $x \in X$
- $x \mapsto \nu(E_x)$ is $\mathcal{A}$ -measurable
- $(\mu \otimes \nu)(E) = \int_X \nu(E_x) d\mu(x)$

We can only expect E_x to be $\mathcal{B}$ -measurable for μ -a.e. x ! Example, add...

[Since E is measurable, there exist $E_k \subseteq \bigcup \mathcal{A} \times \mathcal{B}$ such that $E \subseteq E_k$ and $(\mu \otimes \nu)(E_k \setminus E) \rightarrow 0$. Typical trick: write $E' = \bigcap E_k$, so that $E' = E \sqcup N$ for N negligible. Assume that N_x is negligible for μ -a.e. x (hence measurable by completeness). (**Not** for all! Think of a line in $\mathbb{R}^2$.) Then from previous lem, $E_x = E'_x \setminus N_x$ is $\mathcal{B}$ -measurable for μ -a.e. x . Thus $\nu(E_x) = \nu(E'_x)$ μ -a.e., so other two points go through.

To check N $\mu \otimes \nu$ -negligible implies N_x ν -negligible for μ -a.e. x , choose $N' \in \bigcap \bigcup \mathcal{A} \times \mathcal{B}$ with $(\mu \otimes \nu)(N') = 0$ such that $N' \supseteq N$. Then $\int_X \nu(N'_x) d\mu(x) = 0$, so $\nu(N'_x) = 0$ for μ -a.e. x .]

We now give a formula for integrating $d(\mu \otimes \nu)$.

Thm (Fubini-Tonelli): let $f : X \times Y \rightarrow \mathbb{R} \cup \{\pm\infty\}$ be $\overline{\mathcal{A} \otimes \mathcal{B}}$ -measurable. Assume $f \geq 0$ (Tonelli) or $f \in L^1(\mu \otimes \nu)$ (Fubini). Then

- $y \mapsto f(x, y)$ is $\mathcal{B}$ -measurable for μ -a.e. x (and ν -integrable if $f \in L^1$)
- $x \mapsto \int_Y f(x, y) d\nu(y)$ is $\mathcal{A}$ -measurable
- $\int_{X \times Y} f(x, y) d(\mu \otimes \nu) = \int_X (\int_Y f(x, y) d\nu(y)) d\mu(x)$

[Tonelli: since $f \geq 0$, there exists a sequence of nonnegative simple functions $f_k \nearrow f$. We have shown the result for χ_E with $E \in \overline{\mathcal{A} \otimes \mathcal{B}}$, so by linearity we obtain the result for nonnegative simple functions f_k . Use MCT to pass from f_k to f in all three properties: $[y \mapsto f_k(x, y)] \nearrow [y \mapsto f(x, y)]$ for μ -a.e. x , so by MCT $\int_Y f_k(x, y) d\nu(y) \nearrow \int_Y f(x, y) d\nu(y)$ for μ -a.e. x , and by one more application of MCT we obtain the iterated integral.

Fubini: write $f = f^+ - f^-$ for integrable nonnegative f^+, f^- and apply Tonelli to each.]

Lem: $\mathcal{L}^{d_1+d_2} = \mathcal{L}^{d_1} \otimes \mathcal{L}^{d_2}$.

[Evaluate both sides on rectangles, and apply Caratheodory.]

Counterexamples when assumptions are removed / when formulae only hold μ -a.e.? Add...

§0.5 Signed measures

Defn: let $\mathcal{A} \subseteq 2^X$ be a σ -algebra. A function $\mu : \mathcal{A} \rightarrow \mathbb{R}$ is a (finite) signed measure if $\mu(\emptyset) = 0$ and μ is σ -additive (where all sums converge absolutely).

Examples: the signed measure $\mu^+ - \mu^-$ for a pair of measures μ^+, μ^- , the *finite* signed measure $f\mu$ for $f \in L^1(\mu)$ for some measure μ .

Defn: given a signed measure space $(X, \mathcal{A}, \mu)$, let the total variation $|\mu| : \mathcal{A} \rightarrow [0, \infty]$ be given by

$$|\mu|(E) = \sup_{\sqcup_{k \in \mathbb{N}} E_k = E} \sum_{k \in \mathbb{N}} |\mu(E_k)| \quad (2)$$

The total variation of E is the largest possible sum of absolute values across all countable partitions of E . As a simple example, note $|f\mu| = |f|\mu$. We restrict to *finite* signed measures.

Lem: if $|\mu|(E) > c > 0$, then there exists $E' \subseteq E$ such that $|\mu(E')| > \frac{c}{2}$.

[Take a partition (E_k) with $\sum |\mu(E_k)| > c$. Group the positive and negative contributions to this sum, and call these sets F^+ and F^- . At least one must exceed $\frac{c}{2}$ in measure. This is E' .]

Thm: if μ is a finite signed measure, then $|\mu|$ is a finite *measure* such that $|\mu(E)| \leq |\mu|(E)$.

[$|\mu(E)| \leq |\mu|(E)$ is obvious.

We check $|\mu|$ is σ -additive. Take (E_k) disjoint. Then by taking a refinement,

$$|\mu|(\bigsqcup_k E_k) = \sup_l \sum_l |\mu(F_l)| \quad (3)$$

$$= \sup_l \sum_l \left| \sum_k \mu(F_l \cap E_k) \right| \quad (4)$$

$$\leq \sup_k \sum_k \sum_l |\mu(F_l \cap E_k)| \quad (5)$$

$$= \sum_k |\mu|(E_k) \quad (6)$$

where the sup is over $\bigsqcup_l F_l = \bigsqcup_k E_k$. For the other direction, let $\bigsqcup_l E_{k,l}$ be a partition of E_k . Then $|\mu|(\bigsqcup_k E_k) \geq \sum_{k,l} |\mu(E_{k,l})|$. Taking a sup over l , $|\mu|(\bigsqcup_k E_k) \geq \sum_k |\mu|(E_k)$. This yields σ -additivity.

We check $|\mu|$ is finite. FTSOC $|\mu|(E) = \infty$ for some E . Then by lem, we can iteratively construct a sequence of disjoint (E_k) such that $|\mu(E_k)| > 1$, violating absolute convergence requirement.]

Thm (Hahn-Jordan decomposition): let $(X, \mathcal{A}, \mu)$ be a finite signed measure space. Then there exists a partition $X = P \sqcup N$ where $P, N \in \mathcal{A}$ such that $\mu(E) \geq 0$ for all $E \subseteq P$ and $\mu(E) \leq 0$ for all $E \subseteq N$. Moreover if $\mu_P = \mu(E \cap P)$ and $\mu_N(E) = -\mu(E \cap N)$, then μ_P, μ_N are finite measures such that $\mu_P = \frac{|\mu| + \mu}{2}$ and $\mu_N = \frac{|\mu| - \mu}{2}$. Finally, this decomposition is unique.

[The proof is quite indirect. Intuitively, P is the set with maximal μ -measure. Let $\alpha = \sup_E \mu(E)$.

We have the following overlap result: $|\mu|(E \triangle F) \leq 4 \max(\alpha - \mu(E), \alpha - \mu(F))$.

Now take a sequence $(P_k) \subseteq \mathcal{A}$ such that $\mu(P_k) \nearrow \alpha$. By the overlap result, $|\mu|(P_k \triangle P_l) \leq \varepsilon$ for all sufficiently large k, l , so that $\|\chi_{P_k} - \chi_{P_l}\|_{L^1(|\mu|)} \leq \varepsilon$. Thus (χ_{P_k}) forms a Cauchy sequence in $L^1(|\mu|)$, so by completeness it has a limit $f \in L^1(|\mu|)$. By the partial converse to DCT, there exists a subsequence $\chi_{P_{k_i}} \rightarrow f$ $|\mu|$ -a.e., so $f = \chi_P$.

Now we claim $\mu(P) = \alpha$. Note $\mu(P) = \mu(P_k) - \mu(P \setminus P_k)$. Since $\mu(P_k) \rightarrow \alpha$ and $\mu(P \setminus P_k) = \mu(P \triangle P_k) \rightarrow 0$, we obtain $\mu(P) = \alpha$.

Set $N = X \setminus P$. Note $E \subseteq P$ implies $\mu(E) = \mu(P) - \mu(P \setminus E) \geq 0$, since P attains the sup. Similarly if $E \subseteq N$ then $\mu(E) = \mu(P \sqcup E) - \mu(P) \leq 0$ since P attains the sup.]

Counterexample when signed measure μ is not finite? add...

§0.6 Hilbert spaces

Mostly a repeat of linear algebra. Recall definition of $\mathbb{C}$ - or $\mathbb{R}$ -Hilbert space: complete inner product space. A Hilbert space lets us discuss geometry, like orthogonality.

Examples: square integrable functions $L^2(X, \mathcal{A}, \mu)$, where $\langle f, g \rangle = \int_X f \bar{g} d\mu$ ($\mathbb{C}$ -Hilbert space), square summable sequences $\ell^2(\mathbb{N}, \mathbb{C}) = \{(x_k)_{k \in \mathbb{N}} : \sum_k |x_k|^2 < \infty\}$ with $\langle x, y \rangle = \sum_k x_k \bar{y}_k$, any *closed subspace* of a Hilbert space.

Lem: Cauchy-Schwarz and the triangle inequality.

[Cauchy-Schwarz implies triangle inequality, because $|x + y|^2 = |x|^2 + |y|^2 + 2\langle x, y \rangle \leq |x|^2 + |y|^2 + 2|x||y| = (|x| + |y|)^2$, so $|x + y| \leq |x| + |y|$.

For Cauchy-Schwarz, consider the trivial $|x - ty|^2 \geq 0$. Expanding (in real case), $|y|^2 t^2 - 2\langle x, y \rangle t + |x|^2 \geq 0$. Discriminant must be ≤ 0 : $|\langle x, y \rangle| \leq |x||y|$. ($\mathbb{C}$ -case? add...)]

Defn: given $S \subseteq H$, $S^\perp$ is the set of vectors orthogonal to every element of S .

(Note $S^\perp$ is a closed subspace, even if S is an arbitrary set. Indeed, it equals $\bigcap_{s \in S} \ker[x \mapsto \langle x, s \rangle]$. The kernel of a linear map is a subspace. Since $x \mapsto \langle x, s \rangle$ is *bounded*, it is continuous, so the preimage of the closed set 0 is also closed.)

Lem (projection onto V): if H is a Hilbert space and V is a *closed subspace*, then there exists a unique linear 1-Lipschitz map $\pi_V : H \rightarrow V$ such that $x - \pi_V(x) \in V^\perp$ for all $x \in H$.

[Intuitively, $\pi_V(x)$ should be closest point in V to x . How to find it? Given x , let $L = \inf_{v \in V} |x - v|$. Take infimizing sequence $(y_k) \subseteq V$. Claim (y_k) has a limit in V . Since V closed implies V complete, just check (y_k) is Cauchy via parallelogram law: $|y_k - y_l|^2 = 2|x - y_k|^2 + 2|x - y_l|^2 - 4|x - (y_k + y_l)|^2 \leq 2(L^2 + \frac{\varepsilon}{2}) + 2(L^2 + \frac{\varepsilon}{2}) - 4L^2 = \varepsilon$ for k, l sufficiently large. Thus $|x - y| = L$ for some $y \in V$, and we define $\pi_V(x) = y$.

$x - y \in V^\perp$: trick is to use minimality of $|x - y|$: given $v \in V$, this implies $|x - (y + tv)|^2 \geq |x - y|^2$ for all t , which yields $\langle x - y, v \rangle = 0$.

Uniqueness: if $x = y + w = y' + w'$ for $y, y' \in V$ and $w, w' \in V^\perp$, then $y - y' = w' - w \in V \cap V^\perp = \{0\}$.

Linearity: directly check $\alpha x + \beta y - (\alpha \pi_V(x) + \beta \pi_V(y)) \in V^\perp$, so by uniqueness $\pi_V(\alpha x + \beta y) = \alpha \pi_V(x) + \beta \pi_V(y)$.

1-Lipschitz: always true of a projection; if $x = y + w$ and $x' = y' + w'$, then $|\pi_V(x) - \pi_V(x')| = |y - y'| \leq \sqrt{|y - y'|^2 + |w - w'|^2} = |x - x'|$ by Pythagoras.]

Thm (Riesz representation): let H be a Hilbert space and let $T : H \rightarrow \mathbb{C}$ (or $\mathbb{R}$) be a bounded linear functional. Then there exists $v \in H$ such that $Tx = \langle x, v \rangle$ for all $x \in H$.

[*A posteriori*, if $v \neq 0$, then $\ker T = \text{span}\{v\}^\perp$ and $\ker T^\perp = \text{span}\{v\}$. So we can hope to locate v in $\ker T^\perp$. Another key property of v is $Tv = |v|^2$. So let us now find such a v *a priori*.

We use T bounded (hence continuous) to ensure $\ker T = T^{-1}(0)$ closed. This is necessary for the orthogonal decomposition $H = \ker T \oplus \ker T^\perp$.

Suppose $T \neq 0$. Claim $\dim \ker T^\perp = 1$. Given $u, v \in \ker T^\perp \setminus \{0\}$, there exists λ such that $Tu = \lambda Tv$. Thus $T(u - \lambda v) = 0$, so $u - \lambda v \in \ker T$. But $u - \lambda v \in \ker T^\perp$, so $u - \lambda v = 0$!

Now pick $w \in \ker T^\perp$ with $Tw = 1$. Let $v = \frac{w}{|w|^2}$. Then $Tv = |v|^2$. Thus given $x \in H$, write $x = k + \lambda v$ for some $k \in \ker T$, $\lambda \in \mathbb{R}$, and note $Tx = \lambda |v|^2 = \langle \lambda v, v \rangle = \langle x, v \rangle$, as desired!]

Defn: $(v_k)_{k \in \mathbb{N}}$ is an orthonormal system if $\langle v_k, v_l \rangle = \delta_{kl}$. (Not the same as basis!)

Lem: let (v_k) be an ON system. Let $V = \text{span}(v_k)$. Then for all $x \in H$, we have $\pi_V(x) = \sum_k \langle x, v_k \rangle v_k$ and $|\pi_V(x)|^2 = \sum_k |\langle x, v_k \rangle|^2$.

[Show for a finite set (v_k) . To determine $\pi_V(x)$, always check its defining property: note $\langle \pi_V(x) - \sum_k \langle x, v_k \rangle v_k, v_l \rangle = 0$ for all l , so the first entry is zero. Extend to infinite ON systems via partial sums.]

Defn: $(v_k)_{k \in \mathbb{N}}$ is an orthonormal basis if it is an ON system and if $\overline{\text{span}(v_k)} = H$. (The span consists of *finite* linear combinations, hence the need to take the closure.)

Thm: let H be a Hilbert space and let (v_k) be an ON system. Let $V = \overline{\text{span}(v_k)}$. TFAE:

- $V = H$
- $x = \sum_k \langle x, v_k \rangle v_k$
- $|x|^2 = \sum_k |\langle x, v_k \rangle|^2$
- if $\langle x, v_k \rangle = 0$ for all k , then $x = 0$

[Add...]

Thm: let H be a separable $\mathbb{R}$ -Hilbert space. Then H is either isomorphic to $\mathbb{R}^d$ for some d or $\ell^2(\mathbb{N})$.

[We only consider the case when $\dim H = \infty$. Let (v_k) be a countable dense subset of H . By Gram-Schmidt we may assume that (v_k) is an ONB. Consider the natural map $\phi : \ell^2(\mathbb{N}) \rightarrow H$ given by $(a_k) \mapsto \sum_k a_k v_k$. We claim that ϕ is an isomorphism of Hilbert spaces. Add...]

§0.7 Radon–Nikodym

Always working on measurable space $(X, \mathcal{A})$.

Typically apply R-N wrt $\nu = \mathcal{L}^d$ (?).

Defn: $\mu \ll \nu$ (absolutely continuous, a.c.) if $\nu(E) = 0$ implies $\mu(E) = 0$.

For any nonnegative $f \in L^1(\nu)$, $f\nu \ll \nu$.

Defn: $\mu \perp \nu$ (mutually singular) if there exist $E, F \in \mathcal{A}$ s.t. $\text{supp}(\mu) \subseteq E$, $\text{supp}(\nu) \subseteq F$, and $E \cap F = \emptyset$.

Ex: $\mu = \mathcal{L}^1$, $\nu =$ counting meas on $\mathbb{N}$. μ is supported on $\mathbb{R} \setminus \mathbb{N}$, ν is supported on $\mathbb{N}$. They see disjoint subsets of $\mathbb{R}$.

Thm (R-N, v1): if μ, ν measures, μ σ -finite, then there exists μ_{ac}, μ_s s.t. $\mu = \mu_{ac} + \mu_s$ s.t. $\mu_{ac} \ll \nu$ and $\mu_s \perp \nu$. Moreover, decomposition is unique. (Decompose into a.c. and singular part.)

[Existence: WLOG μ finite. Let $\alpha = \sup_{E: \nu(E)=0} \mu(E)$. Intuitively, maximizer should be support of singular part. Take $\mu(E_k) \nearrow \alpha$ with $\nu(E_k) = 0$ and let $N = \bigcup E_k$, so that $\mu(N) = \alpha$ and $\nu(N) = 0$. Define $\mu_s = \mu|_N$ and $\mu_{ac} = \mu|_{N^c}$. STS $\mu_{ac} \ll \nu$. Assume FTSOC $\nu(E) = 0$ but $\mu_{ac}(E) > 0$. Then $\mu(E \setminus N) > 0$, so $\mu(E \cup N) = \mu(E \setminus N) + \mu(N) > \alpha$. We have found $E \cup N$ that is ν -negligible but with μ -measure $> \alpha$, contradiction.]

We now characterize μ_{ac} as $f\nu$. f can be thought of “derivative of μ wrt ν ”; though this will only become apparent in v3.

Thm (R-N, v2): μ, ν measures, μ σ -finite, $\mu \ll \nu$. Then there exists a nonnegative measurable f s.t. $\mu = f\nu$.

[von Neumann’s proof via Riesz: WLOG μ, ν finite. consider the operator $T : L^2(\mu + \nu) \rightarrow \mathbb{R}$ given by $T\varphi = \langle \varphi, 1 \rangle_{L^2(\mu)}$. By C-S, bounded. Thus by Riesz, $T\varphi = \langle \varphi, g \rangle_{L^2(\mu + \nu)}$ for some $g \in L^2(\mu + \nu)$. Rearrange: $\int (1 - g)\varphi d\mu = \int g\varphi d\nu$. Take $\varphi = \frac{1}{1-g}\chi_E$ to deduce $\mu(E) = \int_E \frac{g}{1-g} d\nu$, hence $f = \frac{g}{1-g}$. (Modulo showing $g \in [0, 1]$ μ and ν a.e., add...)]

We now restrict to $(X, \mathcal{A}) = (\mathbb{R}^d, \overline{\mathcal{B}})$ (?) and μ, ν locally finite Borel measures (in particular, Radon measures). We now observe the resemblance to the standard definition of derivative.

Thm (R-N, v3): μ, ν locally finite Borel measures on $\mathbb{R}^d$. Then $\mu = \frac{d\mu}{d\nu}\nu + \mu_s$, where we define

$$\frac{d\mu}{d\nu}(x) = \lim_{r \rightarrow 0} \frac{\mu(B_r(x))}{\nu(B_r(x))} \text{ for } \nu\text{-a.e. } x \quad (7)$$

This Thm reduces to the following two results, which handle singular and a.c. cases separately:

Prop: μ, ν locally finite Borel measures on $\mathbb{R}^d$. If $\mu \perp \nu$, then $\frac{d\mu}{d\nu} = 0$ ν -a.e.

Thm (Lebesgue differentiation): μ locally finite Borel measure on $\mathbb{R}^d$. Then if $f \in L^1(\mu)$, $\lim_{r \rightarrow 0} \int_{B_r(x)} |f(y) - f(x)| d\mu(y) = 0$ μ -a.e. In particular, $\lim_{r \rightarrow 0} \int_{B_r(x)} f d\mu = f(x)$ μ -a.e.

[Why do these imply Thm? By R-N v2, write $\mu = f\nu + \mu_s$. Thus $\frac{\mu(B_r(x))}{\nu(B_r(x))} = \lim_{r \rightarrow 0} \int_{B_r(x)} f d\nu + \frac{d\mu_s}{d\nu}$. Both terms vanish by the above.]

Need two ingredients to prove Prop + Lebesgue differentiation.

Prop: if μ is σ -finite Borel measure on $\mathbb{R}^d$, then μ is outer regular and $C_c^0(\mathbb{R}^d)$ is dense in $L^1(\mu)$.

Thm (Besicovitch covering): if $\mathcal{F}$ is family of balls in $\mathbb{R}^d$ with uniformly bounded radii, there exists a subfamily $\mathcal{F}'$ that still covers all centers in $\mathcal{F}$, but such that each point in $\mathbb{R}^d$ is covered by at most $N(d)$ balls in $\mathcal{F}'$. (“Finitely intersecting subfamily.”)

[Why do these imply Prop? Estimate ν -measure of the set F_λ of “bad x ” using “ λ of room.” Suppose $\text{supp}(\nu) \subseteq E$ with $\mu(E) = 0$. Using outer regularity take $O \supseteq E$ open s.t. $\mu(O \setminus E) \leq \varepsilon$. Let $F_\lambda = \{x \in O : \limsup \frac{\mu(B_r(x))}{\nu(B_r(x))} > \lambda\}$. O open implies for each x there exists $B_r(x) \subseteq O$ with $r \leq 1$ s.t. $\frac{\mu(B_r(x))}{\nu(B_r(x))} > \lambda$. By Besicovitch we can extract a “finitely intersecting” subfamily (B_k) . Note (!) $\nu(F_\lambda) \leq \sum \nu(B_k) \leq \sum \frac{1}{\lambda} \mu(B_k) \leq \frac{N}{\lambda} \mu(O)$. Can replace O with E in definition of F_λ and send $\varepsilon \rightarrow 0$ to obtain $\nu(\tilde{F}_\lambda) = 0$ for all λ , hence $\nu(\text{bad points}) = 0$.]

[Why do these imply Lebesgue differentiation? Lebesgue differentiation is completely obvious for continuous functions. We attempt to use density to upgrade to $L^1(\mu)$. What happens? First, take a “ λ of room.” Pick $\tilde{f}$ continuous and “close to” f in $L^1(\mu)$. By triangle inequality, $\limsup \int |f(y) - f(x)| d\mu(y) \leq |\tilde{f}(x) - f(x)| + \limsup \int |\tilde{f}(y) - f(y)| d\mu(y)$. Thus

$$\mu\{\limsup \int |f(y) - f(x)| d\mu(y) > \lambda\} \leq \mu\{|\tilde{f}(x) - f(x)| > \frac{\lambda}{2}\} \quad (8)$$

$$+ \mu\{\limsup \int |\tilde{f}(y) - f(y)| d\mu(y) > \frac{\lambda}{2}\} \quad (9)$$

Define the L_{weak}^1 norm, which is bounded by L^1 . Then the bound reads $\frac{1}{\lambda^{1/2}} \|\tilde{f} - f\|_{L_w^1} + \frac{1}{\lambda^{1/2}} \|M(\tilde{f} - f)\|_{L_w^1}$. Here we define the maximal function M (really an operator) by $Mg(x) := \sup_{r \rightarrow 0} \int_{B_r(x)} |g(y)| d\mu(y)$. Since $\|\tilde{f} - f\|_{L^1}$ is small, the first term is small. Now invoke the fact that $M : L^1(\mu) \rightarrow L_w^1(\mu)$ is a bounded nonlinear operator to conclude the second term is $\leq \frac{C}{\lambda^{1/2}} \|\tilde{f} - f\|_{L^1}$, hence small. The claim follows.

How to prove boundedness of M ? WTS $\lambda \mu(\{Mf > \lambda\}) \leq C \|f\|_{L^1}$. Let $E_\lambda = \{Mf > \lambda\}$ be our “bad” set. For each $x \in E_\lambda$, pick $B_r(x)$ s.t. $\int_{B_r(x)} |f| d\mu > \lambda \mu(B_r(x))$. By Besicovitch, we can extract a “finitely intersecting” subfamily (B_k) . Thus (!) $\mu(E_\lambda) \leq \sum \mu(B_k) \leq \frac{1}{\lambda} \sum \int_{B_k} |f| d\mu \leq \frac{N}{\lambda} \int_{\mathbb{R}^d} |f| d\mu$. We take $C = N$.]

[Pf of Besicovitch: Assume within bounded domain. To construct $\mathcal{F}'$, use greedy algorithm. At each step, take a ball B of maximal radius, and remove all balls whose centers lie in B . Well, can’t quite do that, since the maximum is not necessarily attained; so take B such that $(1 + \delta) \cdot \text{radius}(B)$ is larger than all other radii. Let $\mathcal{F}' := (B_k) := (B_{r_k}(x_k))$ denote the sequence of balls obtained in this manner. Disjointness result: for $k < h$, the “shrunk” balls $B_{r_k/3}(x_k)$ and $B_{r_h/3}(x_h)$ are disjoint. Indeed, since $k < h$, $r_h \leq (1 + \delta)r_k$; since h was not removed, $|x_h - x_k| \geq r_k$; so $|x_h - x_k| \geq \frac{r_k}{3} + \frac{r_h}{3}$. Since in bounded domain, must have $r_k \rightarrow 0$. Claim every center is covered. Suppose x is not covered, where $B_r(x) \in \mathcal{F}$. Eventually $r_k < \frac{r}{2}$, so we would have picked $B_r(x)$ as stage k , contradiction. Remains to show that any point is covered by $\leq N(d)$ balls in $\mathcal{F}'$. Suppose x belongs to $B'_1, \dots, B'_M$ (chosen in that order). WLOG $B'_M = B_1(0)$, so that $(1 + \delta)r'_i \geq 1$. Since x lies in all, B'_i intersects $B_1(0)$ for all i . Split the balls into “big” and “small”.

Group 1: balls of radius $< \delta^{-1}$. Volume argument: all belong to $B_{1+2\delta^{-1}}(0)$, and when shrunk by a factor of 3, are disjoint. Thus $N_1(d) \leq \mathcal{L}^d(B_{1+2\delta^{-1}}(0)) / \mathcal{L}^d(B_{1/3(1+\delta)}(0))$.

Group 2: balls of radius $\geq \delta^{-1}$. Angle argument: can show that $\widehat{p_i 0 p_j} \geq \theta$ for all i, j . The normalized centers $q_i := \frac{p_i}{|p_i|}$ lie on S^{d-1} are well-separated, so we obtain disjoint balls $(B_{\delta'/2}(p_i))$ in $B_2(0)$. Thus $\leq N_2(d)$ many.]

§0.8 Differentiability

Q: which functions f obey the FTC? (i.e. $f(b) - f(a) = \int_{[a,b]} f' d\mathcal{L}^1$)

Guess: every function that is differentiable a.e.?

Counterexample: the Cantor staircase function.

The precise class is defined at the end of this section. Overview: given a BV function f , we may associate to it a locally finite signed measure μ s.t. $\mu((a, b]) = f(b) - f(a)$. By R-N, $\mu = f' d\mathcal{L} + \mu_s$. If we evaluate both sides on $E = [a, b]$, we obtain $f(b) - f(a) = \int_{[a,b]} f' d\mathcal{L} + \mu_s([a, b])$. The obstruction to FTC lies in the singular part μ_s . (Think of μ induced by the Cantor staircase function; $\mu \perp \mathcal{L}$ with no atoms.) If we restrict further to so-called AC functions, then $\mu_s \equiv 0$ and FTC holds.

Prop: let I be an interval. Then every locally finite Borel measure μ corresponds to an increasing right continuous function $F : I \rightarrow \mathbb{R}$ s.t. $\mu((a, b]) = F(b) - F(a)$ for all $a, b \in I$. (F is the “cdf” of μ .)

$$\{ \text{loc fin Borel measures} \} \leftrightarrow \{ \text{increasing r.c. functions} \} \quad (10)$$

[($\rightarrow$) WLOG $I = \mathbb{R}$. Just define $F(t) = \mu((-\infty, t])$. F is r.c. because $t_k \searrow t$ implies $(-\infty, t_k] \searrow (-\infty, t]$. ($\leftarrow$) Define μ on rectangles by $\mu((a, b]) = F(b) - F(a)$ and use Caratheodory. (Use r.c. to check σ -additivity on the semiring.)]

Thm: let I be an interval. Let $f : I \rightarrow \mathbb{R}$ be increasing. Then f is differentiable $\mathcal{L}^1$ -a.e. and $f \in L^1_{loc}(I)$.

(When we write “loc”, we mean “genuinely L^1 when restricted to any compact.” A constant function is in $L^1_{loc} \setminus L^1$ if $I = \mathbb{R}$.)

[Sketch: a consequence of R-N. WLOG $I = \mathbb{R}$. Suppose f is r.c. Consider the associated measure μ . By R-N v3, $\mu = \frac{d\mu}{d\mathcal{L}} \mathcal{L} + \mu_s$, where $\frac{d\mu}{d\mathcal{L}}(x) = \lim_{r \rightarrow 0} \frac{\mu((x, x+r])}{r}$ $\mathcal{L}$ -a.e. But this is precisely $f'(x)$, so f' exists a.e. (A general f can be made r.c.)]

A fuzzy principle: “the classical derivative is the a.c. part of the distributional derivative”; $\mu = f' \mathcal{L} + \mu_s$. (Fuzzier still: functions $\rightarrow$ measures $\rightarrow$ distributions...)

Easy generalization: bounded variation functions.

Defn: $f : I \rightarrow \mathbb{R}$ is BV if $\sup_{a_1 < \dots < a_n} \sum_{k=1}^{n-1} |f(a_{k+1}) - f(a_k)| < \infty$. (“Total distance traveled, if one were to hike along the function.”) BV_{loc} defined in the obvious way.

Ex: $\sin \frac{1}{x}$ is not BV, $x \sin \frac{1}{x}$ is not BV, but $x^2 \sin \frac{1}{x}$ is BV. The first two are BV_{loc} on any $I \not\ni 0$.

Prop: suppose $f : I \rightarrow \mathbb{R}$ is bounded. Then $f \in BV(I)$ iff $f = f^+ - f^-$ for f^+, f^- bounded and increasing. (“BV = difference of two increasing functions.”)

[($\leftarrow$) Easy: split $|f(a_{k+1}) - f(a_k)| \leq (f^+(a_{k+1}) - f^+(a_k)) + (f^-(a_{k+1}) - f^-(a_k))$. ($\rightarrow$) Tricky: define f^+ and f^- in the obvious way; “total upwards distance” and “total downwards distance” respectively.]

Rmk: α -Hölder continuous does not imply BV; for instance, a sample path of standard Brownian motion is a.s. α -Hölder for any $\alpha < \frac{1}{2}$, but is differentiable almost nowhere, hence not BV. Lipschitz functions (i.e. 1-Hölder) are obviously BV_{loc} .

Our correspondence trivially extends to BV functions (?): if $f \in BV(I)$ is bounded, then it defines a signed measure μ such that $\mu = f'\mathcal{L} + \mu_s$.

$$\{ \text{loc fin signed Borel measures} \} \leftrightarrow \{ \text{BV functions} \} \quad (11)$$

We arrive at the class of functions that obey FTC. The name is uncreative; we call the function absolutely continuous because the induced measure μ will be a.c. wrt $\mathcal{L}$.

Defn: $f : I \rightarrow \mathbb{R}$ is absolutely continuous ($f \in AC(I)$) if for all ε , there exists δ s.t. if $\mathcal{L}(\bigsqcup_{k=1}^n I_k) < \delta$ for a disjoint union of intervals (I_k) , then we can bound the total oscillation of f on these intervals by ε : $\sum_{k=1}^n \text{osc}(f, I_k) < \varepsilon$. (Here $\text{osc}(f, I) := \sup_I f - \inf_I f$.) (If $n = 1$, we recover the definition of continuity.)

It is obvious that Lipschitz implies AC.

Thm: let $f : I \rightarrow \mathbb{R}$ be a function. TFAE:

- $f \in AC_{loc}(I)$
- there exists $g \in L^1_{loc}(I)$ s.t. $f(b) - f(a) = \int_{[a,b]} g' d\mathcal{L}$
- f is differentiable a.e., $f' \in L^1_{loc}$, and the FTC holds

To prove this, we use the following

Lem: $f \in AC(I)$ implies $f \in BV_{loc}$.

Lem: given $f \in BV(I)$, define $f^+(a, b)$ as the positive part of the total variation on (a, b) . Then for $a < b < c$, we have $f^+(a, c) = f^+(a, b) + f^+(b, c)$.

Assuming these, we may prove the Thm.

[(3) $\implies$ (2) is trivial. Upon localization, (2) $\implies$ (1) follows from uniform integrability of the integrable function g . (u.i. can handle disjoint unions of intervals.) (1) $\implies$ (3) is what we are really after; and it ultimately reduces to R-N. We show f^+ and f^- obey FTC, which implies that f obeys FTC. First, from Lem, f is BV_{loc} . Now consider $\psi : b \mapsto f^+(a, b)$ for fixed a . ψ is increasing and in fact AC_{loc} . We claim the corresponding measure μ is a.c. wrt $\mathcal{L}$. Here is where the definition of AC enters: if E is $\mathcal{L}$ -negligible, then E can be covered by a union of disjoint intervals $(I_k = (a_k, b_k])$ with total measure δ , hence by AC condition $\sum_{k=1}^{n-1} \text{osc}(\psi, I_k) < \varepsilon$. Since ψ is increasing, $\text{osc}(\psi, I_k) = \psi(b_k) - \psi(a_k) = \mu(I_k)$. Thus $\sum_{k=1}^{n-1} \mu(I_k) = \mu(E) < \varepsilon$. Hence E is μ -negligible, and $\mu \ll \mathcal{L}$. By R-N, we have $\mu = \psi' \mathcal{L}$ with no singular part, hence FTC.]

Lem (IBP): let $f \in AC_{loc}(\mathbb{R})$ and $\varphi \in \mathcal{C}_c^1(\mathbb{R})$. Then $\int_{\mathbb{R}} f' \varphi = \int_{\mathbb{R}} f \varphi'$.

Thm (Rademacher): suppose $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz. Then f is differentiable a.e. Further, the divergence theorem holds: for any vector field $X \in \mathcal{C}_c^1(\mathbb{R}^d, \mathbb{R}^d)$, $\int_{\mathbb{R}^d} \nabla f \cdot X = - \int_{\mathbb{R}^d} f \text{div}(X)$.

To prove Rademacher, we use the following

Lem: let $\bar{x}$ be a Lebesgue point for $\partial_1 f, \dots, \partial_d f$. (We say $\bar{x}$ is a *Lebesgue point* if the conclusion of Lebesgue differentiation holds at $\bar{x}$: $\int_{B_r(\bar{x})} |g - g(\bar{x})| \rightarrow 0$ as $r \rightarrow 0$.) Then f is differentiable at $\bar{x}$ with differential $df(\bar{x}) = \partial_1 f dx_1 + \dots + \partial_d f dx_d$.

[By translating f and subtracting the linearization df from f , we may assume WLOG $\bar{x} = 0$, $f(0) = 0$, and $\partial_1 f(0) = \dots = \partial_d f(0) = 0$. In this case, WTS $df(0) = 0$, i.e. $f(x) = o(|x|)$ as $x \rightarrow 0$. Consider the

“blow up” $f_\varepsilon(x) = \frac{f(\varepsilon x)}{\varepsilon}$: zoom in at 0 while preserving area. Clearly if f is L -Lipschitz, then so is f_ε . Since the family $\{f_\varepsilon : \varepsilon \in (0, 1]\}$ is equicontinuous, Arzela-Ascoli implies that there exist $\varepsilon_k \searrow 0$ such that $f_{\varepsilon_k} \rightarrow f_0$ locally uniformly in $\mathbb{R}^d$ (i.e. uniformly on any compact). Clearly the limit f_0 satisfies $f_0(0) = 0$ and f_0 L -Lipschitz. We claim $\partial_i f_0 = 0$ for all i a.e., so that $f_0 \equiv 0$. This implies the result, since the convergence $f_{\varepsilon_k} \rightarrow 0$ locally uniformly is equivalent to $df(0) = 0$. Indeed, for any $\varphi \in \mathcal{C}_c^1(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \partial_i f_0 \cdot \varphi = \lim_k \int_{\mathbb{R}^d} \partial_i f_{\varepsilon_k} \cdot \varphi \quad (12)$$

so that $|\int \partial_i f_0 \cdot \varphi| \leq \lim_k \|\varphi\|_\infty \int_{B_1(\bar{x})} |\partial_i f(\varepsilon_k x)| dx$. Intuitively the integral on the RHS is getting smaller, since $f(\varepsilon_k x)$ is getting “flatter”. This is where we use that $\bar{x}$ is a Lebesgue point for $\partial_i f$: $\int_{B_1(\bar{x})} |\partial_i f(\varepsilon x)| dx = \int_{B_\varepsilon(\bar{x})} |\partial_i f(y)| dy$, which tends to zero as $\varepsilon \rightarrow 0$.]

With the lemma, we may prove Rademacher.

[We reduce to the 1-d case. From the lemma (?), ∇f exists a.e. and equals $[\partial_i f]_{i \in [d]}$. (Not clear apriori! It’s true that all partials exist a.e., but this is *not* the same as the gradient existing.) Thus we may split

$$\int_{\mathbb{R}^d} \nabla f \cdot X = \sum \int_{\mathbb{R}^d} \partial_i f \cdot X_i = \sum \int_{\mathbb{R}^{d-1}} \left(\int_{\mathbb{R}} \partial_i f \cdot X_i dx_i \right) dx_{-i} \quad (13)$$

If we could apply FTC to $\partial_i f \cdot X_i$, we could undo our steps:

$$= - \sum \int_{\mathbb{R}^{d-1}} \left(\int_{\mathbb{R}} f \partial_i X_i dx_i \right) dx_{-i} = - \int_{\mathbb{R}^d} f \operatorname{div}(X) \quad (14)$$

To justify FTC, note $x_1 \mapsto f(x_1, \dots, x_d)$ is Lipschitz and hence AC, so $\partial_1 f$ exists for a.e. x . By FTC, $\int_{\mathbb{R}} \partial_1 f \cdot \varphi dx_1 = - \int_{\mathbb{R}} f \cdot \partial_1 \varphi dx_1$. The result follows.]

§9.9 Change of variables

We want to establish the COV formula $\int f(g(x))g'(x)dx = \int f(y)dy$, or more generally, $\int f(T(x))|\det dT|dx = \int f(y)dy$.

Battle cry: *Remember the Jacobian!*

Prop: let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be linear. Then $\mathcal{L}^d(T(E)) = |\det T| \cdot \mathcal{L}^d(E)$.

[Use the characterization of $\mathcal{L}^d$ as the unique translation invariant d -homogeneous Borel measure that assigns volume 1 to the cube. Also need the polar decomposition to get the precise scaling factor $|\det T|$.]

Given a measurable map $T : X \rightarrow Y$ between measurable spaces $(X, \mathcal{A})$ and $(Y, \mathcal{B})$, let $T_\# \mu$ denote the pushforward of μ by T .

The previous Prop can be rephrased as $(T^{-1})_\# \mathcal{L}^d = |\det T| \mathcal{L}^d$ for $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ linear.

Claim: if $f : Y \rightarrow [0, \infty)$ measurable and $T : X \rightarrow Y$ measurable, then $\int_Y f d(T_\# \mu) = \int_X (f \circ T) d\mu$.

[Standard machine. Let $f = \chi_E$. Then $\int_Y f d(T_\# \mu) = (T_\# \mu)(E) := \mu(T^{-1}(E))$ and $\int_X (f \circ T) d\mu = \mu(T^{-1}(E))$. Then extend to general nonnegative measurable f .]

The previous Prop combined with the Claim implies COV for linear maps. Take $f = g\chi_{T^{-1}(E)}$ for any measurable E to find

$$\int_{T^{-1}(E)} g |\det T| d\mathcal{L}^d = \int_E (g \circ T^{-1}) d\mathcal{L}^d \quad (15)$$

Finally we generalize to Lipschitz homeos T .

Thm: given Ω, Ω' open subsets of $\mathbb{R}^d$, suppose $T : \Omega \rightarrow \Omega'$ is bijective and T, T^{-1} locally Lipschitz. Then $\mathcal{L}^d(T(E)) = \int_E |\det dT| d\mathcal{L}^d$. Also, for any measurable $f : \Omega' \rightarrow [0, \infty)$ we have the COV formula

$$\int_{\Omega'} (f \circ T^{-1}) d\mathcal{L}^d = \int_{\Omega} f |\det dT| d\mathcal{L}^d \quad (16)$$

[Assume Ω, Ω' bounded, T, T^{-1} genuinely Lipschitz. Written differently, we must show $\mathcal{L}_T^d = |\det dT| \mathcal{L}^d$. We first show $\mathcal{L}_T^d \leq |\det dT| \mathcal{L}^d$. This comes down to R-N. Since $\mathcal{L}_T^d \ll \mathcal{L}^d$ and since these measures are finite, R-N implies $\mathcal{L}_T^d = \frac{d\mathcal{L}_T^d}{d\mathcal{L}^d} \mathcal{L}^d$, where the density equals $\lim_{r \rightarrow 0} \frac{\mathcal{L}_T^d(B_r(x))}{\mathcal{L}^d(B_r(x))}$ at a.e. x . STS that the limit is $\leq |\det T|$ a.e. This is a Taylor approximation: assume $x = 0$ and $T(0) = 0$, and note $\mathcal{L}_T^d(B_r(0)) = \mathcal{L}^d(T(B_r(0))) = |\det dT_0| \mathcal{L}^d((dT_0^{-1} \circ T)(B_r(0)))$. Now $dT_0^{-1} \circ T$ has Jacobian I_d at 0, so $dT_0^{-1} \circ T(x) - x = o(\|x\|)$ for x near 0. Given $\varepsilon > 0$, we therefore have $\mathcal{L}_T^d(B_r(0)) \leq (1 + \varepsilon) |\det dT_0| \mathcal{L}^d(B_r(0))$ for all sufficiently small r , which implies $\mathcal{L}_T^d \leq |\det dT| \mathcal{L}^d$. Obtain the other direction via T^{-1} . For the integral formulation, use the standard machine.]

As a consequence, we have polar coordinates: $\int_{\mathbb{R}^2} f d\mathcal{L}^2 = \int_0^\infty \int_0^{2\pi} f(r \cos \theta, r \sin \theta) r d\theta dr$.

§0.10 Counterexamples/tricks

See Federico's outline of exam/practice problems/HWs/memorize...

Remember: always check conditions of theorems...

The most interesting part are counterexamples. (Most involve **Cantor set**.)

For each theorem, ask: **when does this fail to hold?** What are examples for/against this theorem? See Princeton Math Generals?

- When checking that μ is a measure, check $\mu(\emptyset) = 0$
- When applying *decreasing* continuity of measure/*decreasing* MCT, check that the first function has finite measure/integral. (Otherwise, $E_n = \mathbb{R} \setminus [-N, N]$ decreases to $\emptyset$, but has limiting measure ∞ , and same for $f_n = \chi_{E_n}$)
- Fat Cantor set has positive measure but empty interior; can also consider $[0, 1]$ minus an open neighborhood of the rationals...
- Typical trick: if E is measurable, then for every ε there exists a covering E_k by rectangles such that $\mathcal{L}^d(E \setminus E_k) \leq \varepsilon$; $E' = \bigcap E_k$ is such that $E = E' \sqcup N$ for N negligible
- To violate Caratheodory extension, consider (not σ -finite) counting measure on rectangles (i.e. just $\infty!$), which can be extended in many ways
- Any set E of positive Lebesgue measure contains a non-measurable set. Suppose $E \subseteq [0, 1]$; take a *single* representative from each equivalence class of $\mathbb{R}/\mathbb{Q}$ that intersects E ; then the *disjoint* translates by $\mathbb{Q} \cap [0, 1]$ cover E ; but the translates are also contained in $[0, 2]$
- Composition of continuous and measurable remains measurable, but composition of measurable and measurable need not be measurable. Ex: let $g = \text{Cantor staircase plus identity}$, let $A \subseteq g(C)$ be non-measurable, let $B = g^{-1}(C)$ negligible (hence measurable).

Define $h = \chi_B \circ g^{-1}$. Then h is $\{0, 1\}$ -valued with $h^{-1}(\{1\}) = (g^{-1})^{-1}(B) = A$, which is non-measurable

- To prove something for all Borel sets, show it for all open sets, and then show $\{E : \text{property holds for } E\}$ is a σ -algebra (see next bullet)
- Continuous injective function sends Borel to Borel (show for open by filling with compacts). (To get an example as in the bullet below, need the notion of an analytic set?)
- Lipschitz injective sends Lebesgue measurable to Lebesgue measurable (previous + negligible maps to negligible; latter not true for Cantor staircase plus identity!) (Injective is necessary: otherwise, consider the projection $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ of the $\mathcal{L}^2$ -negligible $A \times N$, where A is non-Borel and N is negligible; its image is the non-Borel A)
- There exist Lebesgue measurable non-Borel sets. Take the bijective function $f : [0, 1] \rightarrow [0, 2]$ given by Cantor staircase plus the identity. Note C maps to a set of measure 1, take a non-Lebesgue measurable set $E \subseteq [0, 1]$, note $f^{-1}(E)$ is negligible hence Lebesgue measurable. But $f^{-1}(E)$ not Borel, otherwise its image under *strictly increasing* f $f(f^{-1}(E)) = E$ would be Borel
- Egorov fails if E is not finite measure; take $f_k = \chi_{[k, k+1]} \rightarrow 0$ pointwise
- $\chi_{\mathbb{Q}}$ is measurable, but is not the pointwise limit of continuous functions (consider behavior of sequence near $x = 0$)
- Fatou strict inequality: $\liminf$ should “jump” down, like if $f_n = \chi_{[0, 1]}$ when n even, $f_n = \chi_{(1, 2]}$ when n odd
- Fatou false if not nonnegative: $f_n = -\chi_{[n, n+1]}$ has $\liminf 0$
- If σ -finite, just assume finite...
- Tall-and-skinny rectangle has a.e. but not L^1 convergence
- Typewriter has L^1 but not a.e. convergence (but you can pass to a subsequence!)
- L^p spaces are nested if the space is finite
- $f(x) = \frac{1}{x^{1/3}} \vee \frac{1}{x}$ is $L^2(\mathbb{R})$ but not $L^1(\mathbb{R})$
- Always justify interchanges of infinite sums and integrals (MCT/DCT)...
- Always justify upgrade from pointwise to L^1 convergence (DCT)...
- Can pass from L^1 to pointwise via subsequence
- σ -finite needed for product measure to make sense; recall $\mu = 0$ if countable and ∞ otherwise (not σ -finite!); can define $\mu \otimes \mu = 0$ if $E \subseteq (A \times \mathbb{R}) \cup (\mathbb{R} \times B)$ for either A or B countable; then for $D = \{(x, x)\}$, slicing fails
- Product of $\mathcal{L}^{d_1}$ -negligible set and non- $\mathcal{L}^{d_2}$ measurable set is $\mathcal{L}^{d_1+d_2}$ -negligible, hence $\mathcal{L}^{d_1+d_2}$ -measurable; but not $\mathcal{L}^{d_1} \otimes \mathcal{L}^{d_2}$ measurable! Also, it has many non-measurable slices
- Always check conditions of Fubini-Tonelli; if $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$, then since not L^1 , iterated

integrals disagree

- If a subspace V is not *closed*, then $V \oplus V^\perp = H$ does not in general hold?...
- Riesz fails if T is not bounded; given Hilbert basis (e_n) , let $Tx = \sum_n na_n$
- If $f \in L^2(\mathbb{R})$, it can be convenient to consider a truncation f_n of f that is bounded and compactly supported. Then $f_n \in L^1(\mathbb{R})$, and we can approximate it by simple functions
- R-N v3 fails if you replace balls centered at x with half intervals $(x, x+r)$ and $\mu = \nu$ is a delta measure
- To violate σ -finite, consider a weighted counting measure of the form $\mu = \sum f(x)\delta_x$. (A pair of weighted counting measures yields a counterexample to uniform integrability for $\mu \ll \nu$)
- Lebesgue density theorem is useful to study the measure of the boundary of a set
- There do not exist A, B disjoint with $\mathcal{L}(A \cap (a, b)) = \mathcal{L}(B \cap (a, b)) = 0$ for all (a, b) by Lebesgue density
- Whenever you want to construct a function that satisfies the properties $(P_n)_{n \in \mathbb{N}}$, you can naively try the decaying sum $f = \sum_{n \in \mathbb{N}} \frac{1}{2^n} \cdot (\text{template function adapted to } P_n)$.
- Any monotone function has countably many discontinuities. Given a countable set $A = (a_n)$, a right-continuous function with discontinuities exactly on A is $f = \sum_{n \in \mathbb{N}} \frac{1}{2^n} \chi_{[a_n, \infty)}$. This corresponds to the measure $\sum_{n \in \mathbb{N}} \frac{1}{2^n} \delta_{a_n}$
- If g denotes the Cantor staircase, the “slippery Cantor staircase” f has derivative zero a.e. but is *strictly* increasing! To define it, enumerate the intervals in $\mathbb{R}$ with rational endpoints as $\{(a_n, b_n)\}$ and write $f(x) = \sum_{n \in \mathbb{N}} \frac{1}{2^n} g(\frac{x-a_n}{b_n-a_n})$
- Cantor staircase function has derivative zero a.e. but goes from 0 to 1, violating FTC! (Cantor staircase in real life: “I never seem him working, but he always gets stuff done!”)
- Cantor staircase plus the identity sends Cantor set to $[0, 1]$; “blows up” negligible set to have positive measure! (So inverse “shoves” positive measure into negligible set, as used on HW)
- To check $f \in BV$, sometimes it’s easier to write $f = g - h$ for increasing g, h . (This gives an easy proof that product of BV is BV)
- $x^2 \sin(1/x)$ is BV on $[0, 1]$, but $x \sin(1/x)$ is not
- $f \in BV$ implies $f' \in L^1$? Write $f' \mathcal{L} = \mu_f - \mu_s$?...
- Derivative discontinuous? Take $x^2 \sin(1/x)$ at 0. Derivative not L^1_{loc} ? Take $x^2 \sin(1/x^2)$ at 0...
- AC implies BV_{loc} , but not necessarily BV: if the domain is the unbounded set $\mathbb{R}$, then $f(x) = x$ AC but not BV. BV allows arbitrarily wide partitions
- To check $f \in AC$, sometimes it’s easier to check that f obeys FTC

- Product of AC is AC; composition of AC is not necessarily AC ($\sqrt{x}$ and $x^2 \sin^2(1/x)$); product of Lipschitz is not Lipschitz ($x \cdot x = x^2$)
- Don't forget the Jacobian!